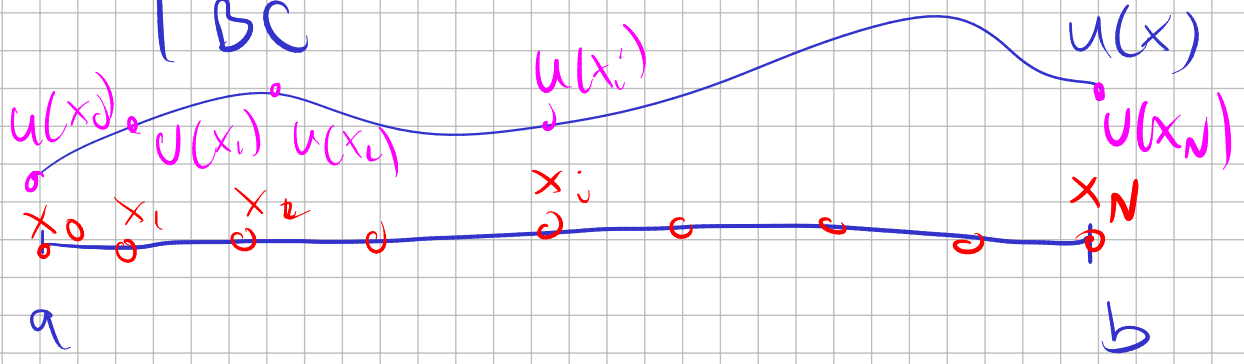
• DIFFERENZE FINITE

- 7-8/4 NO LEZIONE
- 15/4 LEZIONE IN LAB 1
- 5-6/5 NO LEZIONE

→ APPROSIMAZIONE PDE

1D $\Omega = [a,b]$

$$\begin{cases} F(x, u, u_x, u_{xx}, \dots, u^{(k)}) = 0 \\ BC \end{cases}$$



$u_0 \simeq u(x_0)$

$u_1 \simeq u(x_1)$

$u_i \simeq u(x_i)$

$u_N \simeq u(x_N)$

$a = x_0 < x_1 < x_2 < \dots < x_N = b$

$x_{i+1} - x_i = h \quad \forall i = 0, \dots, N-1$

$$\begin{pmatrix} u_0 \\ \vdots \\ u_N \end{pmatrix} = \underline{u} \in \mathbb{R}^{N+1} \quad \text{INCOGNITE}$$

⇒ CERCHIAMO $N+1$ EQUAZIONI

$\partial_x u(x)$

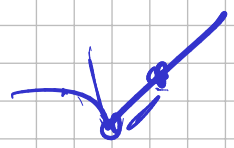
DIFFERENZE DIVISE

$x+h,\ x-h \in [a,b]$

$$\delta_{h,+} u(x) = \frac{u(x+h) - u(x)}{h}$$

$$\delta_{h,-} u(x) = \frac{u(x) - u(x-h)}{h}$$

$$\delta_h u(x) = \frac{u(x+\frac{h}{2}) - u(x-\frac{h}{2})}{h}$$



$$\lim_{h\to 0} \delta_{h,+} u(x) = u'(x)$$

$$\lim_{h\to 0} \delta_{h,-} u(x) = u'(x)$$

$$\lim_{h\to 0} \delta_h u(x) = u'(x)$$

ERRORE?

$$\delta_{h,-} u(x) - u'(x) = \frac{u(x) - \overbrace{u(x-h)}^{\text{TAYLOR}}}{h} - u'(x)$$

$$= \frac{\cancel{u(x)} - [\cancel{u(x)} - h u'(x) + \frac{h^2}{2} u''(\xi)]}{h} - u'(x)$$

$$= \frac{\cancel{h}}{\cancel{h}} u'(x) - \frac{h^{\cancel{2}}}{2} \cdot u''(\xi) - u'(x) = -\frac{h}{2} u''(\xi)$$

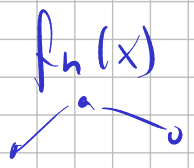
$$|\delta_{h,-} u(x) - u'(x)| \leq \frac{h}{2} \max_{\xi \in [a,b]} |u''(\xi)|$$

$\forall\ x \in [a,b]$

$$|\delta_h u(x) - u'(x)| \le \frac{h^2}{24} \max_{\xi \in [a,b]} |u^{(3)}(\xi)|$$

DEF

ORDINE DI CONVERGENZA



$f \qquad f_h$

$$\lim_{h \to 0} \left(f(x) - f_h(x) \right) = 0$$

DI ORDINE DI ~~CONVERGENZA~~ ACCURATEZZA p

$$|f_h(x) - f(x)| = O(h^p)$$

$x, \; x-h, \; x-2h$

$$f'(x) \approx \overbrace{a_0 f(x) + a_1 f(x-h) + a_2 f(x-2h)}^{\delta_h^* f(x)} \; ?$$

CHI SONO a_0, a_1, a_2 ?

OBIETTIVO $\delta_h^* f \approx f'$ ORDINE PIÙ ALTO POSSIBILE

$\Rightarrow$ APPROX DEL SECOND'ORDINE

TUTTA A SX

DERIVATE + ALTE

2 VOLTE COLPATA 1° ORDINE

$$\delta_h f(x) = \frac{f(x+\frac{h}{2}) - f(x-\frac{h}{2})}{h}$$

$$\delta_h(\delta_h f)(x) = \frac{f(x+h)-f(x)}{h^2} - \frac{f(x)-f(x-h)}{h^2}$$

$$= \frac{f(x+h) - 2f(x) + f(x-h)}{h^2} \quad \leftarrow !!$$

$$\delta_h^2 f(x) := \delta_h(\delta_h f)(x)$$

$$\left|\delta_h^2 f(x) - f''(x)\right| \leq \frac{h^2}{12} \max_{\xi \in [a,b]} \left|f^{(4)}(\xi)\right|$$

$$-u''(x) = f(x) \qquad x \in [a,b]$$

$$u(a) = u(b) = 0$$

$$-u''(x_i) \approx -\frac{u(x_{i+1}) - 2u(x_i) + u(x_{i-1})}{h^2}$$

$$-\frac{u_{i+1} - 2u_i + u_{i-1}}{h^2} = f(x_i) \qquad \underbrace{i = 1, \dots, N-1}_{N-1 \text{ eq}}$$

DIFF. FINITA

$$\left.\begin{aligned} u_0 &= 0 \\ u_N &= 0 \end{aligned}\right\} 2 \text{ eq BC}$$

$N+1$ equazioni per $N+1$ incognite

$$\underbrace{-\frac{1}{h^2}\begin{bmatrix} -2 & 1 & & & 0 \\ 1 & -2 & 1 & & \\ & 1 & -2 & 1 & \\ & & \ddots & \ddots & \ddots \\ 0 & & & 1 & -2 \end{bmatrix}}_{=:A \approx -D^2} \underbrace{\begin{bmatrix} u_1 \\ u_2 \\ \vdots \\ u_{N-1} \end{bmatrix}}_{:=U} = \underbrace{\begin{bmatrix} f(x_1) \\ \vdots \\ f(x_{N-1}) \end{bmatrix}}_{=:F}$$

SYS LINEARE

$$\underline{\underline{A}}\,\underline{u} = \underline{F}$$

$u(x_0) = \alpha$

$u(x_N) = \beta$

$$\begin{array}{c} \to \\ \to \\ \\ \\ \\ \to \end{array} \begin{bmatrix} 1 & 0 & \cdots & & & 0 \\ -\frac{1}{h^2} & \frac{2}{h^2} & -\frac{1}{h^2} & 0 & \cdots & \\ 0 & -\frac{1}{h^2} & \frac{2}{h^2} & -\frac{1}{h^2} & 0 & \cdots \\ \vdots & \ddots & \ddots & \ddots & \ddots & 0 \\ 0 & \cdots & 0 & -\frac{1}{h^2} & \frac{2}{h^2} & \frac{1}{h^2} \\ 0 & \cdots & & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} u_0 \\ u_1 \\ \vdots \\ u_{N-1} \\ u_N \end{bmatrix} = \begin{bmatrix} \alpha \\ f(x_1) \\ \vdots \\ f(x_{N-1}) \\ \beta \end{bmatrix}$$

$$\partial_x u(a) \simeq \frac{U_1 - U_0}{h} = \alpha$$

$$\begin{bmatrix} -\frac{1}{h} & \frac{1}{h} & 0 & \cdots & 0 \\ -\frac{1}{h^2} & \frac{2}{h^2} & -\frac{1}{h^2} & 0 & \cdots \\ 0 & -\frac{1}{h^2} & \frac{2}{h^2} & -\frac{1}{h^2} & 0 \\ \vdots & \ddots & \ddots & \ddots & 0 \\ 0 & \cdots & 0 & -\frac{1}{h^2} & \frac{2}{h^2} \quad \frac{1}{h^2} \\ 0 & \cdots & & 0 & 0 \quad 1 \end{bmatrix} \begin{bmatrix} U_0 \\ U_1 \\ \vdots \\ U_{N-1} \\ U_N \end{bmatrix} = \begin{bmatrix} \alpha \\ f(x_1) \\ \vdots \\ f(x_{N-1}) \\ \beta \end{bmatrix}$$

$$h_{i+1/2} = x_{i+1} - x_i \neq x_i - x_{i-1} =: h_{i-1/2}$$

ERRORE DI TRONCAMENTO $\quad -u'' - f = 0$

$$-f(x) - \frac{u(x+h) - 2u(x) + u(x-h)}{h^2}$$

DATO DALLA FORMA DISCRETA CHE VOGLIAMO RISOLVERE A CUI SOSTITUIAMO LA SOLUZ ESATTA

$$T(x) := -f(x) - \frac{u(x+h) - 2u(x) + u(x-h)}{h^2}$$

$$= \underbrace{-f(x) - u''(x)}_{=0} + \frac{h^2}{24}\left(u^{(4)}(\xi) + u^{(4)}(\xi)\right)$$

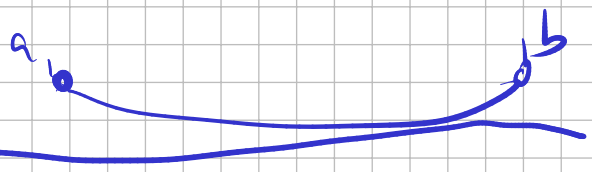
$$|T(x)| \leq \frac{h^2}{12} \max_{\xi \in [a,b]} |u^{(4)}(\xi)|$$

PRINCIPIO DEL MASSIMO

$-u'' = f \Rightarrow$ se $f \leq 0$

PRINCIPIO MASSIMO DISCRETO

$\Rightarrow \max_{x \in [a,b]} u(x) = \max(u(a), u(b))$



$-u'' = f \qquad \Omega = [0,1]$

$N+1$ pti

$$\boxed{\begin{aligned} u(0) &= 0 \\ u(1) &= 0 \end{aligned}}$$

$$\boxed{\underline{\underline{A}} \cdot \underline{U} = \underline{F}} \leq 0 \quad \forall\, i = 0, \dots, N+1$$

$$\Rightarrow \max_{i=1,\dots,N-1} u_i \leq \max\{u_0, u_N\}$$

<u>Dim</u> PER ASSURDO $\exists\, n \quad 1 \leq n \leq N-1$

$\underbrace{v_n > v_0}$ OPPURE $\underbrace{v_n > v_N}$ $\qquad v_n = \max_{i=0,\dots,N} v_i$

$$0 \geq (Av)_n = -\frac{1}{h^2}\left(v_{n+1} - 2v_n + v_{n-1}\right)$$

$$\Rightarrow \frac{v_{n+1} + v_{n-1}}{2} \geq \frac{\cancel{2}v_n}{\cancel{2}} \qquad \boxed{v_n \leq \frac{v_{n+1} + v_{n-1}}{2}}$$

$v_n = v_{n+1} = v_{n-1} = v_0 = v_N \Rightarrow$ ASSURDO.

TEORIA (STIMA A PRIORI DELL'ERRORE)

$$|u(x_i) - u_i| \le \frac{h^2}{24}(b-a)^2 \cdot \max_{\xi \in [a,b]} |u^{(4)}(\xi)|$$

DIM $M = \max_{\xi \in [a,b]} |T(x)|$ $\qquad T(x) := f(x) + \delta_h^2 u(x)$

$$= f(x) + \frac{u(x+h) - 2u(x) + u(x-h)}{h^2}$$

$$\boxed{T(x_i)} = \underbrace{f(x_i)}_{= -\frac{u_{i+1} - 2u_i + u_{i-1}}{h^2}} + \frac{u(x_{i+1}) - 2u(x_i) + u(x_{i-1})}{h^2} \qquad \forall i = 1, \dots, N-1$$

$$(*) = -\frac{u_{i+1} - 2u_i + u_{i-1}}{h^2} + \frac{u(x_{i+1}) - 2u(x_i) + u(x_{i-1})}{h^2}$$

$$e_i := u(x_i) - u_i$$

$$(*) = \boxed{\frac{e_{i+1} - 2e_i + e_{i-1}}{h^2}} \qquad i = 1, \dots, N-1$$

$$e_0 = e_N = 0$$

$$\phi(x) = \frac{M}{2}\left(x - \frac{a+b}{2}\right)^2 \qquad M = \max_{\xi \in [a,b]} |T(x)|$$

$$v_i := e_i + \phi(x_i)$$

$$(Av)_i = -\frac{1}{h^2}(v_{i+1} - 2v_i + v_{i-1}) = -\frac{1}{h^2}(e_{i+1} - 2e_i + e_{i-1}$$

$$+ \phi(x_{i+1}) - 2\phi(x_i) + \phi(x_{i-1})\Big)$$

$$= -T(x_i) - \frac{M}{2h^2}\left[\left(x_i + h - \frac{a+b}{2}\right)^2 - 2\left(x_i - \frac{a+b}{2}\right)^2 + \left(x_i - h - \frac{a+b}{2}\right)^2\right]$$

$$\left(x_i^2 + 2hx_i + h^2 - 2x_i\frac{a+b}{2} + 2h\frac{a+b}{2} + \left(\frac{a+b}{2}\right)^2\right)$$

$$2h^2$$

$$= -T(x_i) - \frac{M}{2h^2}2h^2 = -T(x_i) - \overset{\overset{0}{\uparrow}}{M} \leq 0$$

$$V_i \leq \max\{V_0, V_N\}$$

$$V_0 = e_0 + \phi(x_0) = 0 + \frac{M}{2}\left(a - \frac{a+b}{2}\right)^2$$

$$= \frac{M}{2}\left(\frac{b-a}{2}\right)^2 = \frac{M}{8}(b-a)^2$$

$$V_N = \underset{\overset{\shortparallel}{0}}{e_N} + \overset{\phi(b)}{\phi(x_N)} = \frac{M}{2}\left(b - \frac{a+b}{2}\right)^2$$

$$= \frac{M}{8}(b-a)^2$$

$$V_i \leq \frac{M}{8}(b-a)^2$$

$$e_i = u(x_i) - u_i =$$

$$\| $$

$$v_i - \underbrace{\phi(x_i)}_{\geq 0} \leq v_i \leq \frac{M}{8}(b-a)^2$$

$$\leq \frac{h^2}{24}\underbrace{(b-a)^2} \cdot \max_{\xi \in [a,b]} |u^{(4)}(\xi)|$$

$$\boxed{-e_i} \leq \ldots$$

$$v_i := -e_i + \phi(x_i) \quad \Rightarrow \quad |e_i| \leq \qquad \boxtimes$$

$$\begin{cases} Au = F \\ M > \lambda_i(A) > \varepsilon \\ u \leq F \cdot \|A\| \end{cases}$$

$$\|u\|_2 \leq \underbrace{\|A^{-1}\|_2} \cdot \|F\|_2$$